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QUESTION PAPER

Name of the Examination: WINTER 2022-2023 – CAT-2

Course Code: CSE3001 **Course Title:** Agile Development Process

Set number: I **Date of Exam:** 03/04/2023 (AN) (G12)

Duration: 90 min **Total Marks:** 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. How can an organization effectively envision and prioritize its goals and objectives in an agile framework to ensure alignment and adaptability in a rapidly changing business environment? **(10M)**

Q2. Assume a company that provides home cleaning services wants to expand its business to include commercial cleaning services. Craft a compelling vision statement for your company's expansion into the commercial cleaning services market, that aligns with our Agile principles and values in any two popular vision statement formats. **(10M)**

Q3. How can pair programming be effectively implemented in an Agile development environment to improve code quality, enhance team collaboration, and accelerate knowledge sharing and skills development? **(10M)**

Q4. Do software development teams determine whether FDD or TDD is most appropriate for a project based on their project requirements, team composition, and product goals in an agile development environment? Justify. **(10M)**

Q5. Assume you are planning for a product release with a feature backlog as shown in the table below. Also consider the dependencies between features, and the story point size of each feature as given in the table. Logical connectivity between features can be ignored for the given scenario.

Product Backlog	Dependency	Story Point Size	Priority	Sprint Mapping	Status (Filled/to be refined)
Feature A	dependency on Feature P	5	high		
Feature B	No dependency	3	low		
Feature C	No dependency	7	medium		
Feature D	dependency on Feature N	4	high		
Feature E	dependency on Feature C	5	medium		
Feature F	dependency on Feature M	8	medium		
Feature G	dependency on Feature C	6	low		
Feature H	dependency on Feature A	3	low		

Feature I	No dependency	4	medium		
Feature J	No dependency	3	low		
Feature K	dependency on Feature A	5	high		
Feature L	dependency on Feature M	3	low		
Feature M	No dependency	6	medium		
Feature N	dependency on Feature B	4	high		
Feature O	No dependency	4	high		
Feature P	dependency on Feature F	3	medium		
Feature Q	dependency on Feature H	6	low		
Feature R	dependency on Feature N	5	low		

Sprint size **2 Weeks**

Team's velocity **20 story points**

i. Calculate the number of sprints required and do sprint mapping considering a Fixed scope release planning scenario and plan for a feasible release date. **(5M)**

ii. Considering a fixed-date release planning scenario, and analyzing the feasibility, update the status of each feature to filled/to be refined. (Assuming you are planning to start the next week and the release date is 15th June 2023). **(5M)**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	3	2	-	2	10
Q2	3	3	2	-	2	10
Q3	4	4	1	-	-	10
Q4	4	4	3,5	-	-	10
Q5	3	3	3	-	2	10

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QUESTION PAPER

Name of the Examination: WINTER 2022-2023 – CAT-2

Course Code: CSE3001 **Course Title:** Agile Development Process

Set number: 4 **Date of Exam:** 30/03/2023 (FN) (D1)

Duration: 90 min **Total Marks:** 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. Craft a clear and inspiring vision statement using any two procedures for the development of mobile app that encompasses an Agile approach, and the app can be used by anybody to study foreign languages called Duolingo. **(10M)**

Q2. As a product owner you are planning a new product of mobile to cover the business in the local market of India. Prepare a project road map which will notify the business process, and also it incorporates the changes from the customer which makes the product development more efficient. **(10M)**

Q3. Outline the importance of pair programming. How the pair programming and refactoring helps in Agile development? **(10M)**

Q4. Illustrate the principle of Test-Driven Development with the help of a suitable project in Agile development. **(10M)**

Q5. Assume you are planning for a release with a feature backlog as shown in the table below. Also consider the dependencies between features, and the story point size of each feature as given in the table. Logical connectivity between features can be ignored for the given scenario.

Product Backlog	Dependency	Story Point Size	Priority	Sprint Mapping	Status (Filled/to be refined)
Feature A	dependency on Feature P	5	Low		
Feature B	No dependency	3	High		
Feature C	No dependency	5	Medium		
Feature D	dependency on Feature N	4	Low		
Feature E	dependency on Feature P	2	Low		
Feature F	No dependency	5	Low		
Feature G	No dependency	6	High		
Feature H	dependency on Feature C	6	Low		

Feature I	No dependency	4	medium		
Feature J	dependency on Feature R	5	High		
Feature K	dependency on Feature P	7	High		
Feature L	dependency on Feature A	6	High		
Feature M	No dependency	3	medium		
Feature N	No dependency	3	Low		
Feature O	dependency on Feature H	3	High		
Feature P	dependency on Feature H	4	medium		
Feature Q	dependency on Feature N	5	High		
Feature R	dependency on Feature F	7	high		

Sprint size **2 Weeks**

Team's velocity **20 story points**

i. Calculate the number of sprints required and do sprint mapping considering a Fixed scope release planning scenario and plan for a feasible release date. **(5M)**

ii. Considering a fixed-date release planning scenario, and analyzing the feasibility, update the status of each feature to filled/to be refined. (Assuming you are planning to start the next week and the release date is 15th June 2023). **(5M)**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	3	2	-	2	10
Q2	3	3	2	-	2	10
Q3	4	4	1,3	-	-	10
Q4	4	4	2,5	-	-	10
Q5	3	3	3	-	2	10